

Hydraulic Distribution Pipeline Analysis Report

Evaluasi Diameter Pipa Distribusi Air Bersih — Perumahan Citra Srie Pradita

Darcy-Weisbach / Swamee-Jain hydraulic analysis, pump sizing, exergy (Gouy-Stodola) evaluation, and Monte Carlo uncertainty quantification, generated by the hydraulic-analysis config-driven toolkit.

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1. Introduction & Background

This report evaluates pipe-diameter choices for a residential water distribution network, comparing a narrow (1/2 inch) supply pipe against a wide (4 inch) distribution pipe under an identical design flow rate. The analysis was originally carried out as a standalone calculation for the Citra Srie Pradita housing estate and has since been rebuilt as a reusable, tested, config-driven engineering package so that new pipe sizes, fluids, or operating conditions can be evaluated by editing a YAML configuration file rather than rewriting calculations.

2. Methodology

Major (friction) head loss is computed with the Darcy-Weisbach equation, $h_f = f (L/D)(v^2/2g)$, where the Darcy friction factor f is obtained from the Hagen-Poiseuille relation ($f = 64/Re$) in laminar flow ($Re < 2300$), or the explicit Swamee-Jain (1976) approximation in turbulent flow:

$$f = 0.25 / [\log_{10}(\epsilon/(3.7D) + 5.74/Re^{0.9})]^2$$

Minor losses from fittings (elbows, valves, entrance/exit) are added via the K-factor method, $h_{\text{minor}} = K(v^2/2g)$. Required pump shaft power is $P_{\text{shaft}} = \rho g Q h_{\text{total}} / (\eta_{\text{pump}} \eta_{\text{motor}})$, where h_{total} includes both the friction/minor losses and any static lift or required delivery head. Frictional irreversibility is quantified via the Gouy-Stodola theorem: the exergy destroyed by friction equals $\rho g Q h_f$ (the friction-loss portion only — static lift is reversible, useful work, not destroyed exergy). All inputs are validated against physical plausibility bounds (Poka-Yoke) before any calculation runs, and resulting velocities are checked against the SNI 03-6481-2000 recommended range of 0.9–2.0 m/s.

3. Scenario Comparison

Table 1 compares all scenarios defined in `configs/scenario_config.yaml`, run through the identical calculation pipeline.

Scenario	D (mm)	Q (L/s)	v (m/s)	Re	h _f (m)	Δp (Pa)	P _{shaft} (W)	Exergy (W)
half_inch_baseline	12.7	0.500	3.947	49,977	137.0000	1,339,938.3	992.55	669.969
four_inch_baseline	101.6	0.500	0.062	6,247	0.0078	76.2	0.06	0.038

half_inch_baseline: ⚠ v = 3.947 m/s exceeds SNI maximum (2.0 m/s). Risk of water hammer.

four_inch_baseline: ⚠ v = 0.062 m/s is very low. Risk of sediment deposition. SNI recommends ≥ 0.9 m/s.

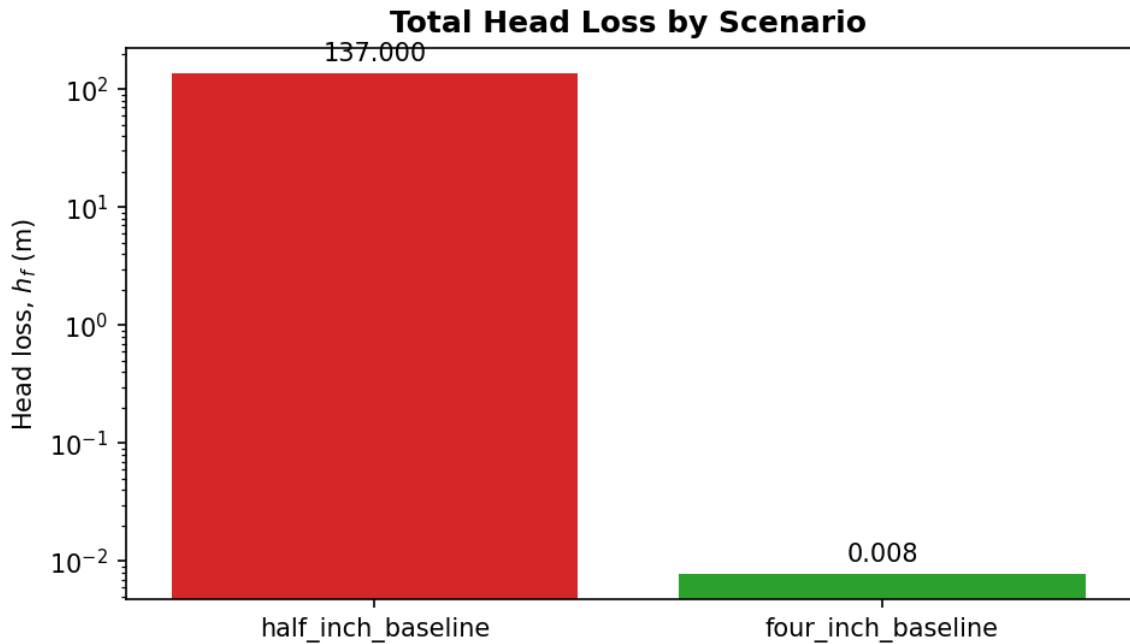


Figure 1. Total head loss by scenario (log scale).

The narrow (1/2 inch) pipe produces approximately 17,593× higher head loss than the wide (4 inch) pipe at the same flow rate — consistent with the Darcy-Weisbach equation's strong (roughly D^{-5}) sensitivity to diameter in the turbulent regime.

4. Diameter Sensitivity Analysis

Sweeping diameter continuously (per the sensitivity block in `scenario_config.yaml`) shows head loss falling sharply as diameter increases, with diminishing returns beyond roughly 3 inches for this flow rate.

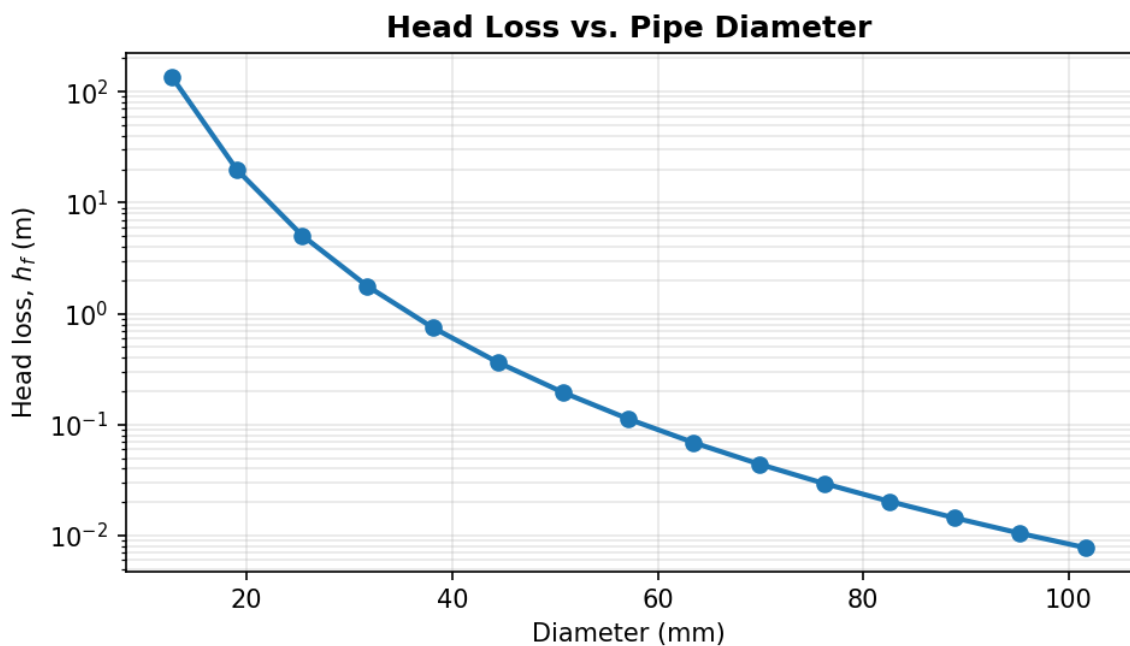


Figure 2. Head loss vs. pipe diameter (log-log).

5. Pressure Profile Along the Pipeline

Assuming a uniform pipe (constant diameter and roughness along its length), pressure declines approximately linearly with distance under the friction-loss contribution, with discrete step-drops at fitting locations.

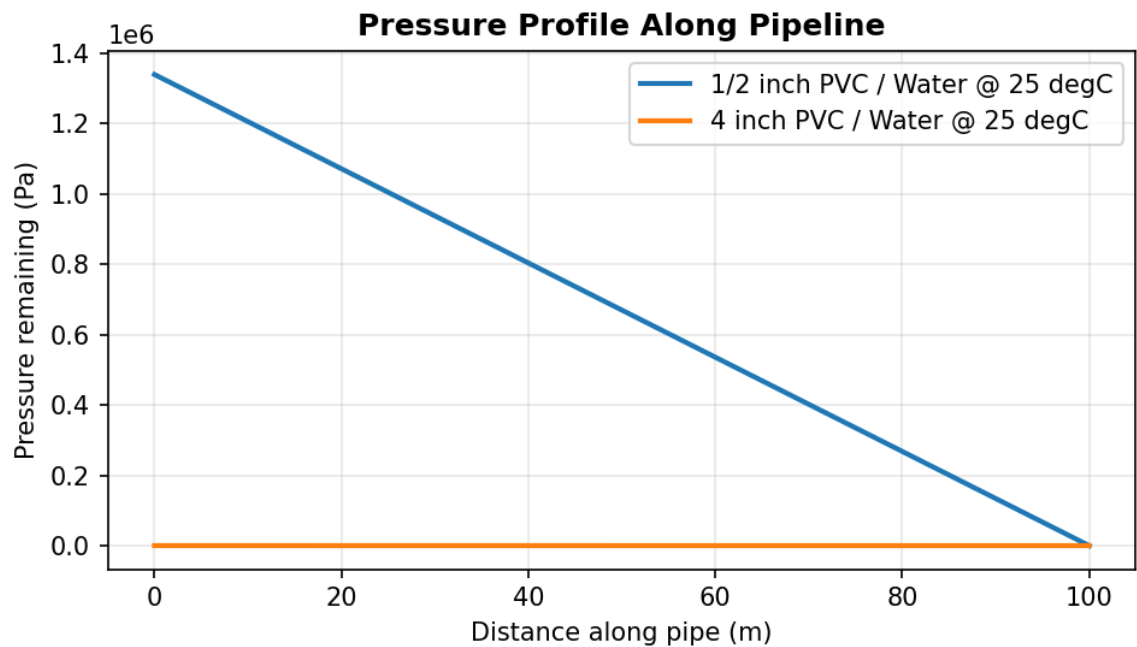


Figure 3. Pressure remaining vs. distance along the pipe, by scenario.

6. Exergy & Energy Balance Analysis

The Gouy-Stodola theorem attributes head loss not just to a pressure deficit but to irrecoverable work potential lost to entropy generation. Figure 4 breaks down required shaft power into motor losses, pump losses, exergy destroyed to friction, and any useful work delivered (static lift). In this analysis, no static lift was configured ($\text{static_head_m} = 0$), so the entire hydraulic power output is destroyed to friction by construction — in a real system with a water tower or required delivery pressure, that component would appear as useful work.

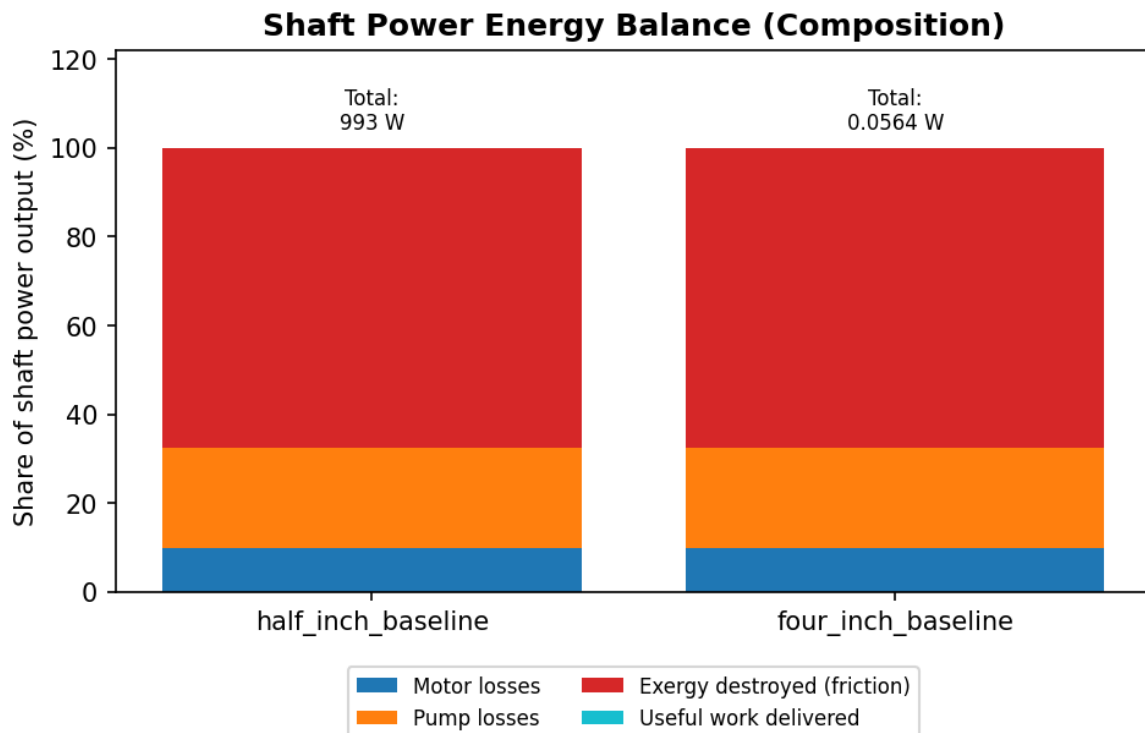


Figure 4. Shaft power energy balance by scenario.

7. Uncertainty Analysis (Monte Carlo)

Flow rate and pipe roughness were sampled per the monte_carlo block in scenario_config.yaml (2000 trials) for the 4 inch PVC / Water @ 25 degC scenario, propagating input uncertainty through the full calculation chain rather than relying on a single point estimate.

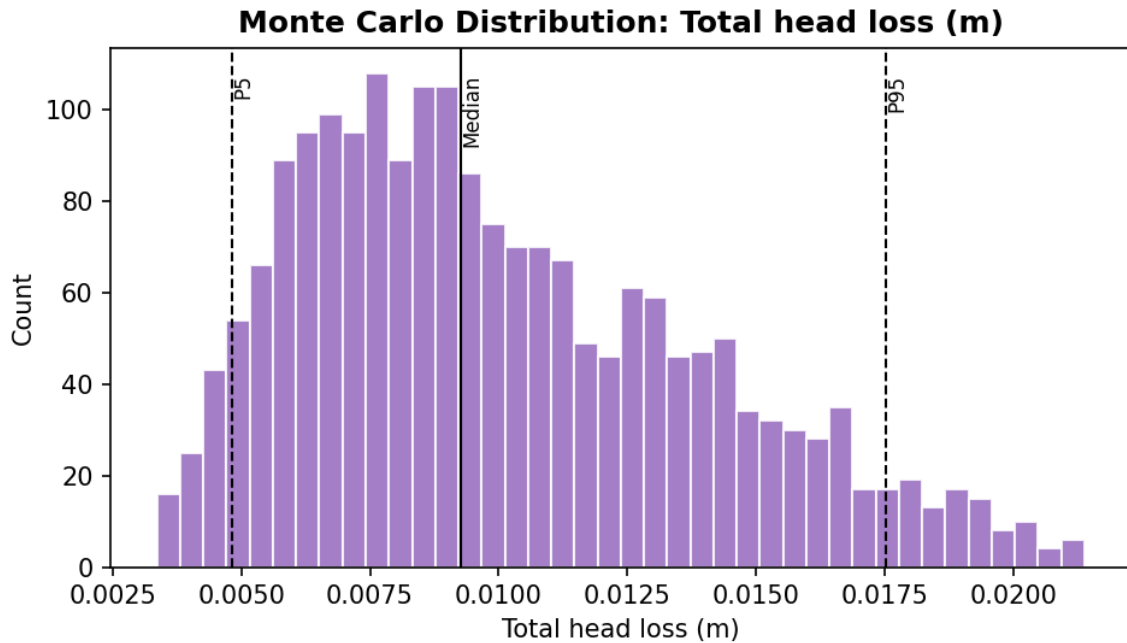


Figure 5. Monte Carlo distribution of total head loss, with P5/median/P95 marked.

Table 2. Monte Carlo summary statistics.

Statistic	total_loss_m	pressure_drop_Pa	shaft_power_W	exergy_destroyed_W
mean	0.01007	98.51	0.09001	0.06076
std	0.003888	38.03	0.05441	0.03673
p05	0.004821	47.15	0.0265	0.01789
median	0.009264	90.61	0.07413	0.05004
p95	0.01753	171.4	0.2016	0.1361

8. Lean / Poka-Yoke Integration

Poka-Yoke (mistake-proofing): every calculation validates diameter, length, flow rate, fluid properties, and pump efficiencies before running — implausible inputs raise a clear error rather than silently producing a wrong number. Resulting velocities are also checked against the SNI 03-6481-2000 recommended range, as shown under Table 1.

Muda (waste): Figure 4 quantifies waste as exergy destroyed to friction. Figure 6 breaks the largest contributor (half_inch_baseline) down by source — friction vs. individual fittings — so improvement effort can target whichever single source dominates, rather than treating head loss as one undifferentiated number.

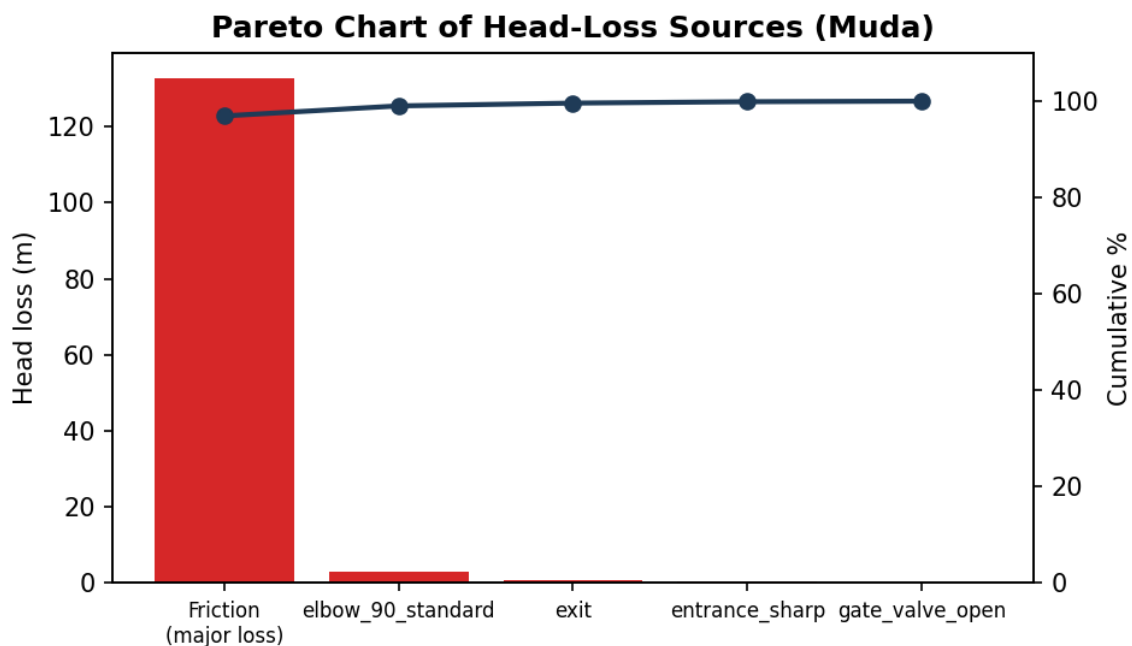


Figure 6. Pareto chart of head-loss sources for half_inch_baseline.

Mura (unevenness): comparing scenarios side by side (Table 1) surfaces load imbalance directly — in this analysis the two scenarios sit at opposite extremes of utilization (one far above, one far below the SNI-recommended velocity band), which is itself the Mura signal: capacity is unevenly matched to demand across the two pipe choices.

Muri (overburden): pump shaft power vs. rated capacity, where a rated capacity is configured:

- half_inch_baseline: 992.55 W required vs. 750.00 W rated (132%) — $\triangle$ Pump overloaded (Muri): operating at 132% of rated capacity (992.55 W vs 750.00 W rated). Risk of premature failure — select a larger pump or reduce demand.
- four_inch_baseline: 0.06 W required vs. 1.00 W rated (6%) — within safe operating margin.

NPSH (cavitation): available suction head vs. required, where suction-side parameters are configured:

- half_inch_baseline: NPSHa = 8.24 m vs. NPSHr = 9.00 m (92% margin) — $\triangle$ Cavitation risk: NPSH available (8.24 m) is below NPSH required (9.00 m) — 92% margin. Expect vapor bubble formation/collapse at the impeller, causing erosion, noise, and performance loss.

- four_inch_baseline: NPSHa = 11.99 m vs. NPSHr = 3.00 m (400% margin) — comfortable margin.

9. Lifecycle Cost Analysis (LCCA)

Translating the hydraulic comparison into financial terms: CAPEX (pipe material/installation) against the present value of OPEX (electricity to overcome friction) over the analysis horizon (20 years, 7.0% discount rate). None of the cost inputs below are real market prices — they are the illustrative placeholders in configs/economics_config.yaml, included to demonstrate the calculation; replace them with real supplier quotes and tariffs before using this for an actual decision. This section presents factual present-value arithmetic only and is not financial advice.

Scenario	D (mm)	Unit Cost (\$/m)	CAPEX (\$)	Annual OPEX (Yr 1, \$)	PV of OPEX (\$)	Total Lifecycle Cost (\$)
half_inch_baseline	12.7	5.00	500.00	1,304.21	13,816.78	14,316.78
four_inch_baseline	101.6	28.00	2,800.00	0.07	0.79	2,800.79

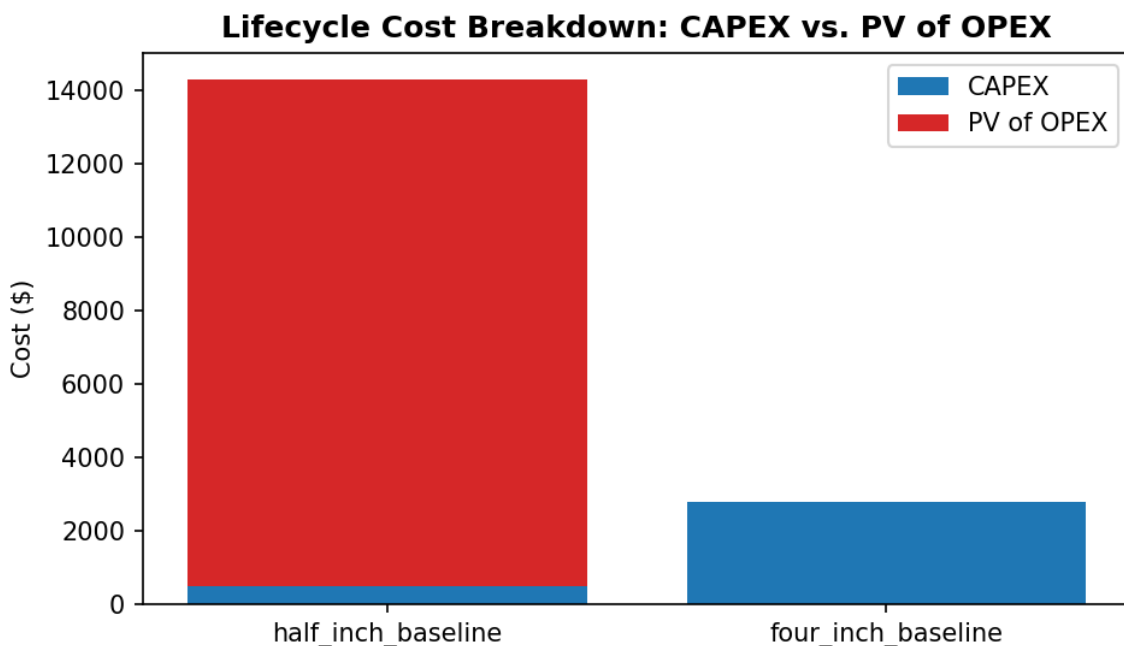


Figure 7. Lifecycle cost breakdown: CAPEX vs. present value of OPEX.

Under these assumptions, four_inch_baseline has the lowest total lifecycle cost (\$2,800.79) despite a higher upfront CAPEX than half_inch_baseline (\$14,316.78 total) — illustrating why CAPEX-only pipe selection can be misleading once operating costs are included.

10. Conclusions & Recommendations

For the evaluated flow rate, the narrow (1/2 inch) pipe is hydraulically impractical, producing both excessive head loss and a velocity far outside the SNI-recommended range. The wide (4 inch) pipe is comfortably within recommended limits at this flow rate; the diameter sensitivity analysis (Section 4) can guide selection of an intermediate size if minimizing material/installation cost is also a design objective. The lifecycle cost analysis (Section 9) reinforces this from a financial angle: the narrow pipe's lower CAPEX is overwhelmed by its OPEX over the analysis horizon. Sensitivity, Monte Carlo, and LCCA results (Sections 4, 7, and 9) should be re-run whenever demand or cost assumptions change — all are config-driven and require no code changes to update.

References

- Swamee, P.K. & Jain, A.K. (1976). Explicit equations for pipe flow problems. *Journal of the Hydraulics Division, ASCE*, 102(5), 657–664.
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